PROBABILITY THEORY
MTM2592-Gr:2

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NORMAL DISTRIBUTION

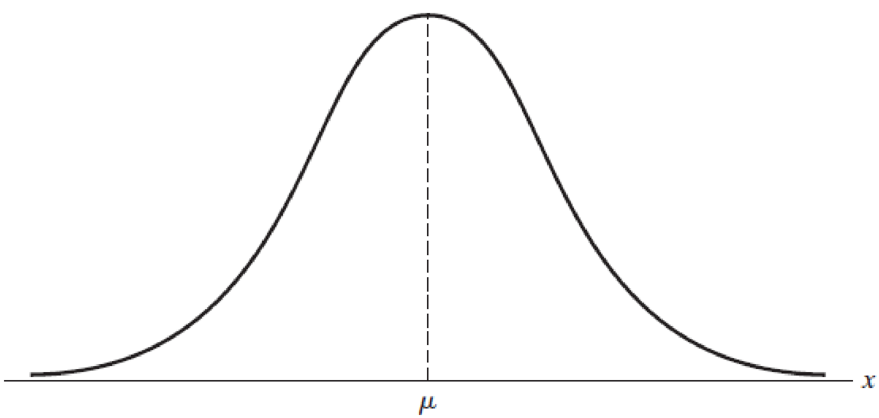
DEFINITION **NORMAL DISTRIBUTION.** *A random variable X has a **normal distribution** and it is referred to as a normal random variable if and only if its probability density is given by*

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

where $\sigma > 0$.

The notation used here shows explicitly that the two parameters of the normal distribution are μ and σ. It remains to be shown, however, that the parameter μ is, in fact, $E(X)$ and that the parameter σ is, in fact, the square root of $\text{var}(X)$, where X is a random variable having the normal distribution with these two parameters.

First, though, let us show that the formula of Definition can serve as a probability density. Since the values of $f(x;\mu,\sigma)$ are evidently positive as long as $\sigma > 0$,

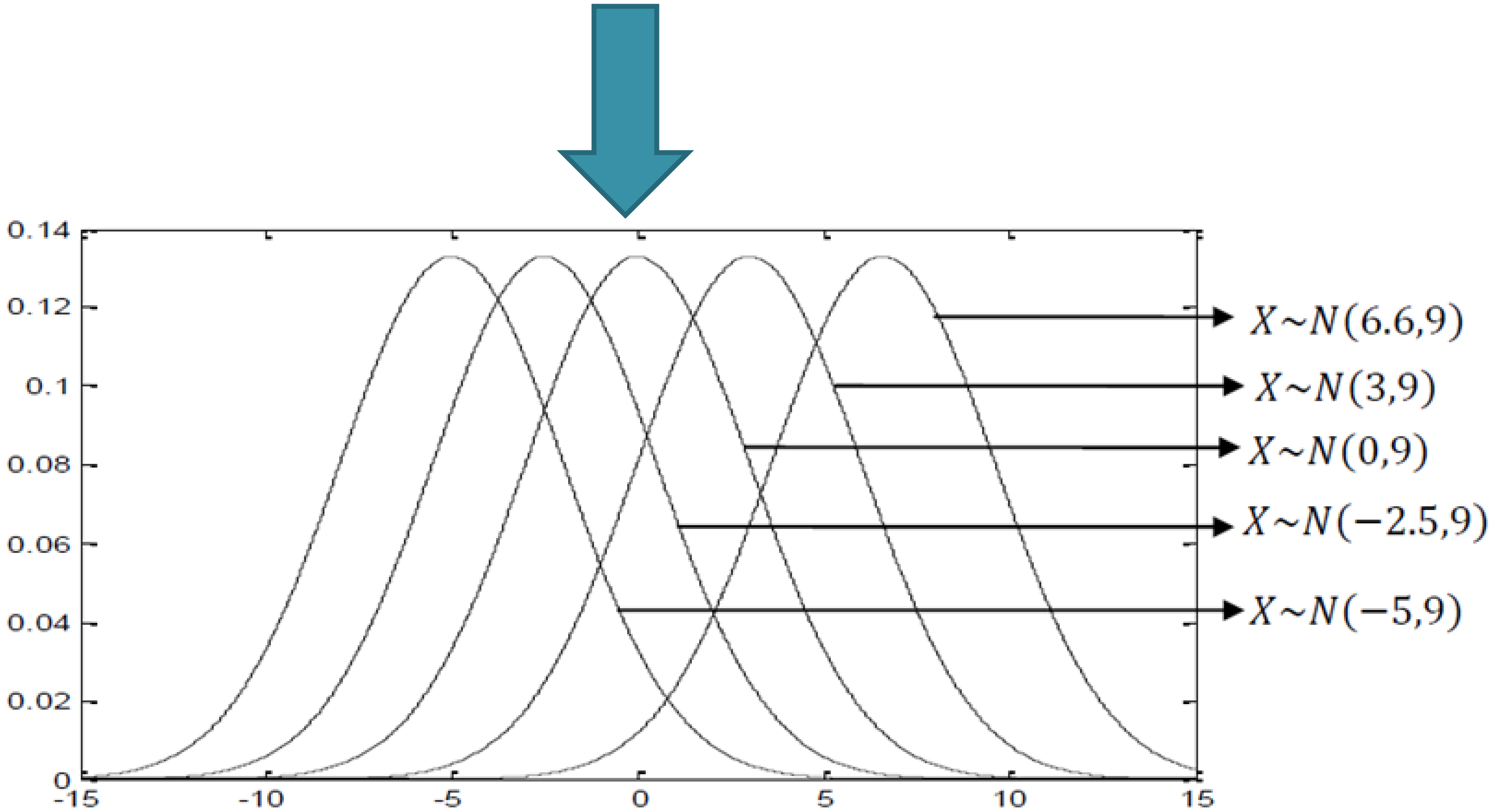


Graph of normal distribution.

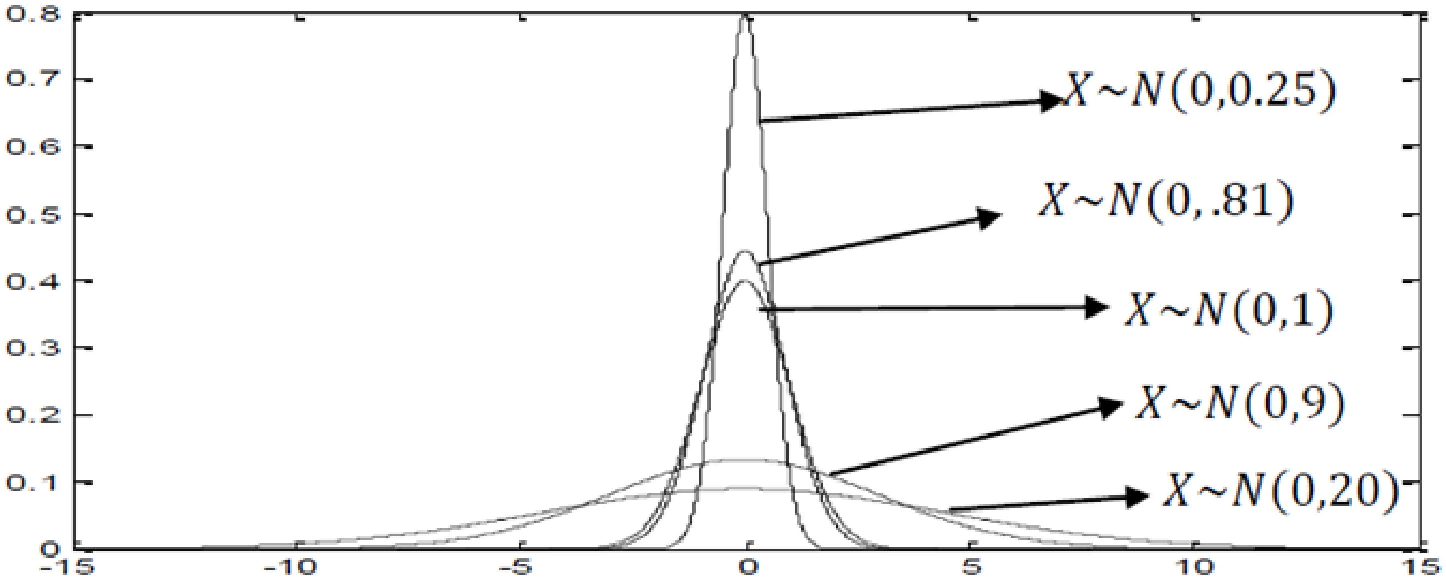
If X has a normal distribution, we denote it as

$$X \sim N\left(\mu, \sigma^2\right)$$

Graph of probability density functions for normal distribution of random variables with the same variance (σ^2=9) but different means;



Graph of probability density functions for normal distribution random variables with the same mean (μ=0) but different variances



Properties of Normal Distribution

1) The area under the normal distribution probability density function is 1.

$$\text{Let show} \quad A = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1$$

$$\text{For this } z = \frac{x-\mu}{\sigma} \Rightarrow dx = \sigma dz$$

$$A = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz$$

$$A^2 = \frac{1}{2\pi}\left(\int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz\right)\left(\int_{-\infty}^{\infty} e^{-\frac{1}{2}u^2} du\right) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{2}\left(z^2+u^2\right)} dzdu$$

$$\text{By using polar coordinates, } z = \rho\cos\theta,\ u = \rho\sin\theta$$

$$dzdu = \rho d\rho d\theta,\ z^2 + u^2 = \rho^2,$$

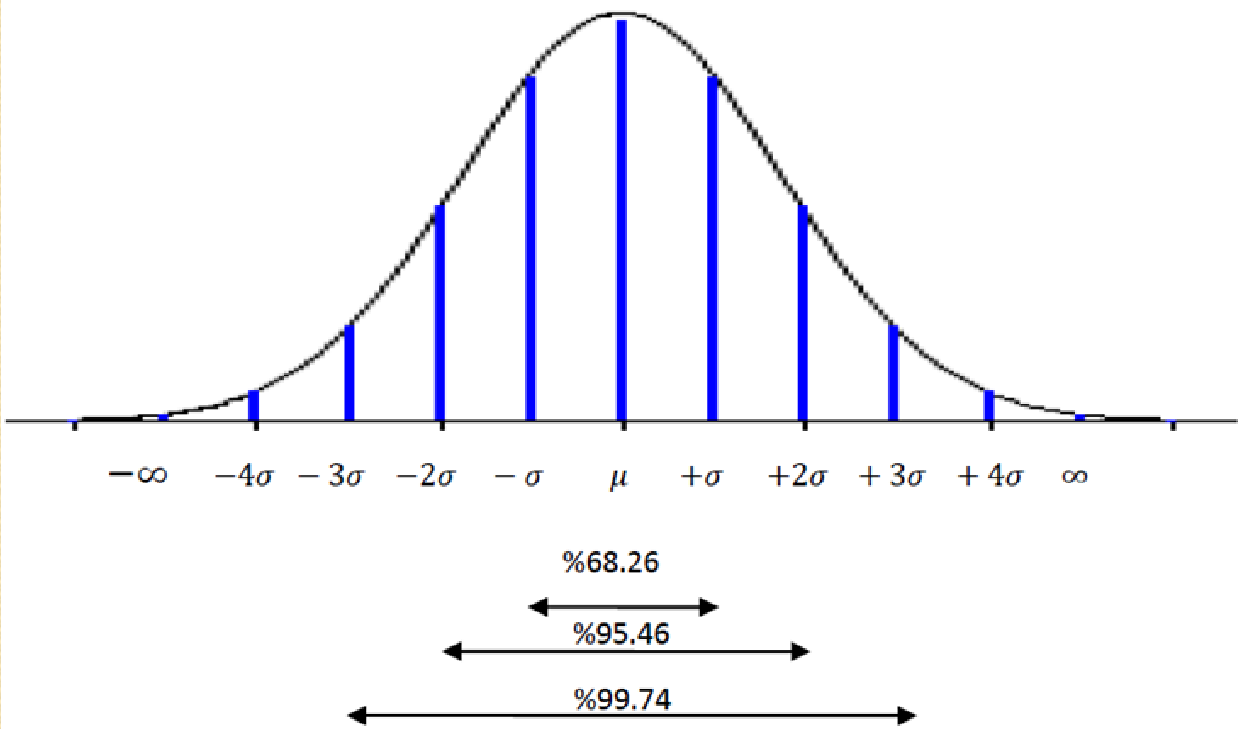
$$A^2 = \frac{1}{2\pi}\int_0^{2\pi} d\theta \int_0^{\infty} \rho e^{-\frac{1}{2}\rho^2} d\rho$$

$$= \frac{1}{2\pi}.2\pi\int_0^{\infty} \rho e^{-\frac{1}{2}\rho^2} d\rho$$

$$A^2 = \int_0^{\infty} \rho e^{-\frac{1}{2}\rho^2} d\rho = 1$$

2) The normal distribution is symmetrical to the mean

$$\int_{-\infty}^{\mu} f(x)\,dx = \int_{\mu}^{+\infty} f(x)\,dx = \frac{1}{2}$$



$68.26\% \Rightarrow \mu \pm 1\sigma$

$95.46\% \Rightarrow \mu \pm 2\sigma$

$99.74\% \Rightarrow \mu \pm 3\sigma$

3) The cumulative function for the normal distribution

$$F(x) = \Phi(x) = P(X \leq x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

this function gives the probability of being less than a certain x value. By using cumulative function;

$$P(X < \infty) = 1$$

$$P(X < \mu) = 0.5$$

$$P(X < \mu + \sigma) = 0.8413$$

$$P(X < \mu + 2\sigma) = 0.9772$$

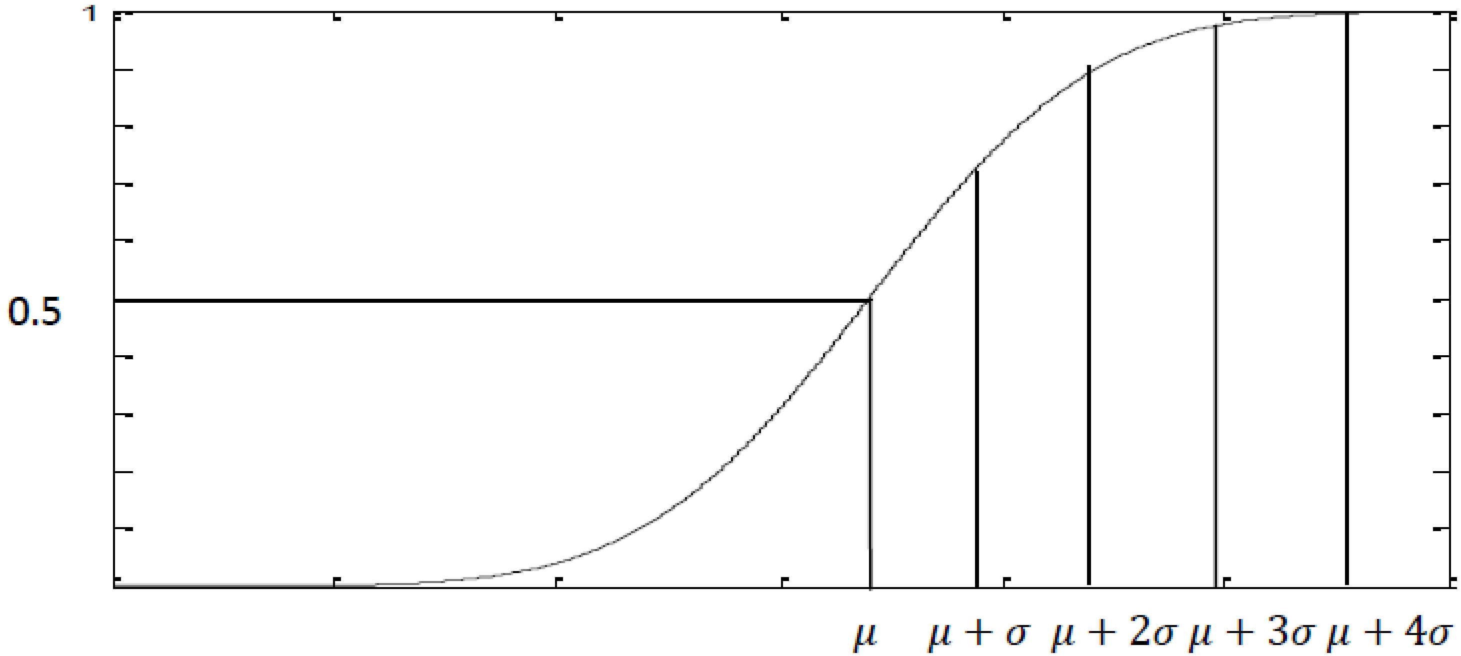
$$P(X < \mu + 3\sigma) = 0.9987$$

$$P(X \le a) = \int_{-\infty}^{a} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = F(a)$$

$$P(X \ge a) = \int_{a}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1 - F(a)$$

$$P(a \le X \le b) = \int_{a}^{b} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = F(b) - F(a)$$

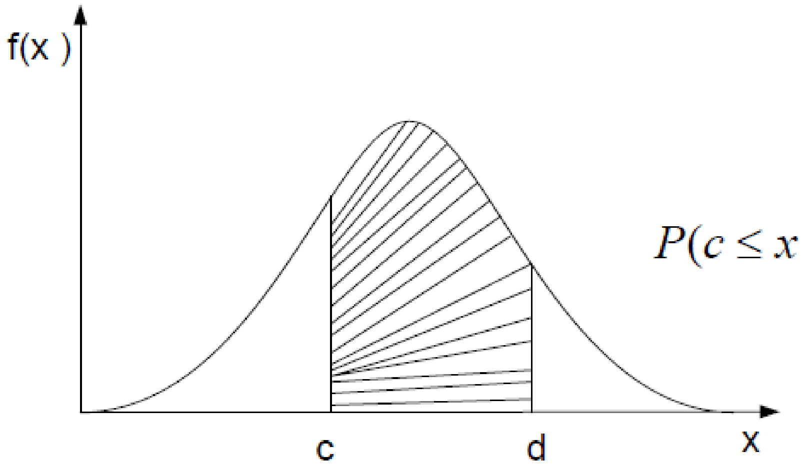
The graph of cumulative function;



f) Normal distribution has a maximum at $x = \mu$

$$Y = f(x) = \frac{1}{\sigma\sqrt{2\pi}}$$

Probability Calculation In Normal Distribution



$$P(c \le x \le d) = \int_c^d f(x)dx = ?$$

$$P(-\infty \le x \le \infty) = \int_{-\infty}^{\infty} f(x)dx = 1$$

Moment Generating Function of Normal Distribution

$$M(t)=E[e^{tX}]=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{+\infty}e^{xt}.e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx \qquad u=\frac{x-\mu}{\sigma}$$

$$x=\sigma u+\mu$$

$$=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{+\infty}e^{t(\mu+u\sigma)}.e^{-\frac{1}{2}u^2}\sigma du=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}e^{tu\sigma}.e^{t\mu}.e^{-\frac{1}{2}u^2}du \qquad dx=\sigma du$$

$$ut\sigma-\frac{1}{2}u^2=-\frac{1}{2}\left(u^2-2ut\sigma\right)=-\frac{1}{2}(u-t\sigma)^2+\frac{1}{2}t^2\sigma^2$$

If we use its identity, we find:

$$M(t)=\frac{1}{\sqrt{2\pi}}e^{t\mu+\frac{1}{2}t^2\sigma^2}\int_{-\infty}^{+\infty}e^{-\frac{1}{2}(u-t\sigma)^2}du$$

$$v=u-t\sigma \qquad \int_{-\infty}^{+\infty}e^{-\frac{1}{2}v^2}dv=\sqrt{2\pi}$$

$$M(t)=\frac{1}{\sqrt{2\pi}}e^{t\mu+\frac{1}{2}t^2\sigma^2}\int_{-\infty}^{+\infty}e^{-\frac{1}{2}v^2}dv \Rightarrow \boxed{M(t)=e^{\mu t+\frac{1}{2}t^2\sigma^2}}$$

$$M'(0)=\mu$$
$$M''(0)=\mu^2+\sigma^2$$
$$E(X)=\mu$$
$$Var(X)=\left(\mu^2+\sigma^2\right)-\mu^2=\sigma^2$$

Expected Value and Variance Of The Normal Distribution

$$E(X)=\int_{-\infty}^{+\infty} x.f(x).dx=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{+\infty} x.e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

$$\frac{x-\mu}{\sigma}=u \qquad x=\sigma u+\mu \qquad dx=\sigma du$$

$$E(X)=\frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^{+\infty}(\sigma u+\mu).e^{-\frac{1}{2}u^2}\sigma du$$

$$E(X)=\frac{\sigma}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}u.e^{-\frac{1}{2}u^2}du+\frac{\mu}{\sqrt{2\pi}}\int_{-\infty}^{+\infty}e^{-\frac{1}{2}u^2}du$$

$$\int_{-\infty}^{+\infty}e^{-\frac{1}{2}u^2}du=\sqrt{2\pi}$$

$$\int_{-\infty}^{+\infty}ue^{-\frac{1}{2}u^2}du=-e^{-\frac{1}{2}u^2}\Big|_{-\infty}^{+\infty}=0$$

$$\boxed{E(X)=\mu.\frac{1}{\sqrt{2\pi}}\sqrt{2\pi}=\mu}$$

From the definition of variance for continuous random variables

$$E(X-E(X))^2 = \int_{-\infty}^{+\infty} (x-\mu)^2 .f(x).dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{+\infty} (x-\mu)^2 .e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\frac{x-\mu}{\sigma} = u$$

$$\sigma_x{}^2 = \frac{1}{\sigma\sqrt{2\pi}} \sigma^2 \int_{-\infty}^{+\infty} u^2 .e^{-\frac{1}{2}u^2} .\sigma du = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} u.\left(ue^{-\frac{1}{2}u^2} \right).du$$

$$= \frac{\sigma^2}{\sqrt{2\pi}} \left\{ -u.e^{-\frac{1}{2}u^2} \Big|_{-\infty}^{\infty} + \int_{-\infty}^{+\infty} e^{-\frac{1}{2}u^2} du \right\}$$

$$u.e^{-\frac{1}{2}u^2} \Big|_{-\infty}^{\infty} = 0 \quad \text{ve} \quad \int_{-\infty}^{+\infty} e^{-\frac{1}{2}u^2} du = \sqrt{2\pi} \quad \Rightarrow \quad \boxed{Var(X) = \sigma^2}$$

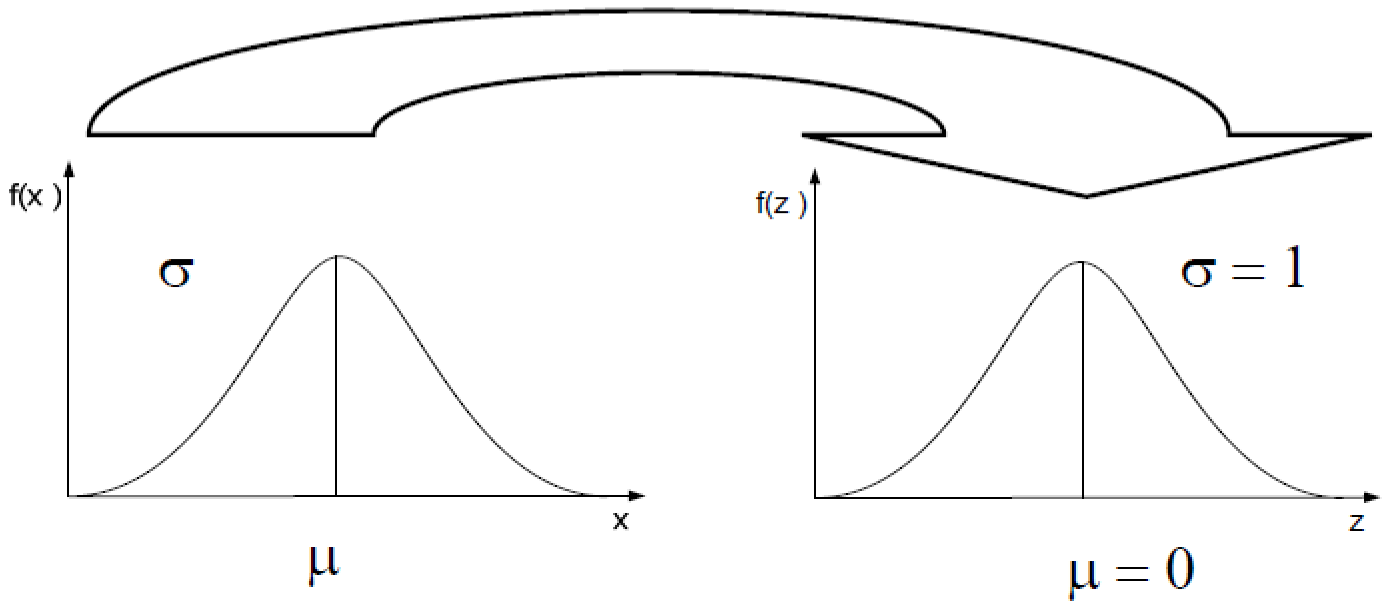
STANDARD NORMAL DISTRIBUTION

The Normal distribution with μ=0 and σ=1 is called the standard normal distribution;

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, -\infty < z < \infty$$

$$z = \frac{x - \mu}{\sigma}$$

- $X \sim N(\mu, \sigma^2)$
- $Z \sim N(0, 1)$



- The curve shown by the standard normal distribution is symmetrical with respect to the f (z) axis.
- The maximum of this curve is the point $\frac{1}{\sqrt{2\pi}}$
- The area under the curve is 1

$$Z \sim N(0,1) \quad a, b \in \mathbb{R}$$ a<b

$$X \sim N(\mu, \sigma)$$

The probability that the random variable is in the range [a, b]

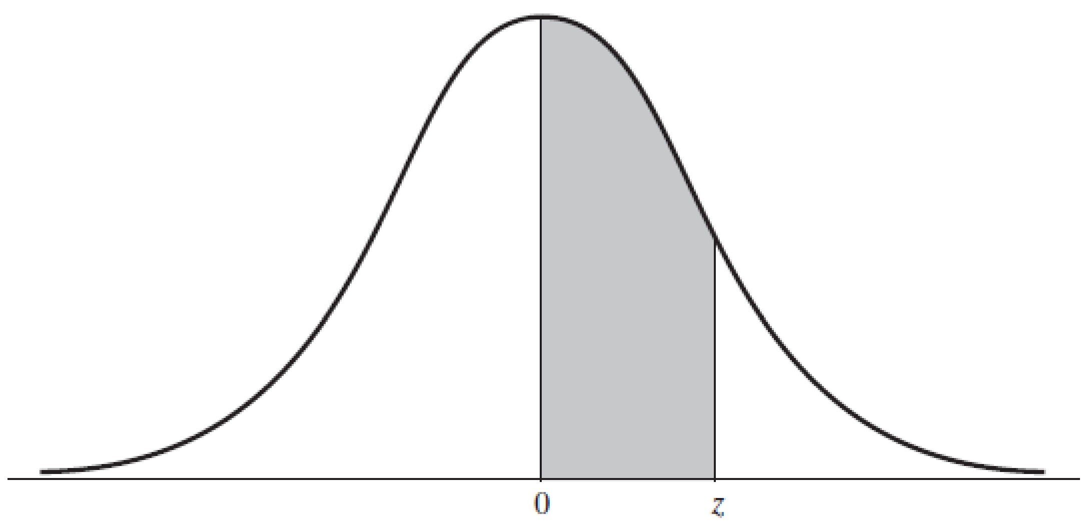
$$P(a < X < b) = P\left(\underbrace{\frac{a-\mu}{\sigma}}_{z_1} < z < \underbrace{\frac{b-\mu}{\sigma}}_{z_2}\right) = \int_{z_1}^{z_2} \frac{1}{\sqrt{2\pi}} . e^{-\frac{1}{2}z^2} dz$$

However, finding this integral is time consuming and difficult.
For this reason, standard normal tables are created.

The entries in standard normal distribution table, represented by the shaded area of Figure , are the values of

$$\int_0^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}\,dx$$

that is, the probabilities that a random variable having the standard normal distribution will take on a value on the interval from 0 to z, for $z = 0.00, 0.01, 0.02, \ldots, 3.08$, and 3.09 and also $z = 4.0$, $z = 5.0$, and $z = 6.0$. By virtue of the symmetry of the normal distribution about its mean, it is unnecessary to extend the table to negative values of z.



Tabulated areas under the standard normal distribution.

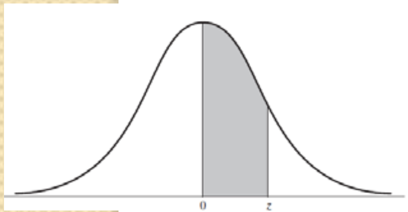


Table III: Standard Normal Distribution

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4988
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Cumulative probability function of Z;

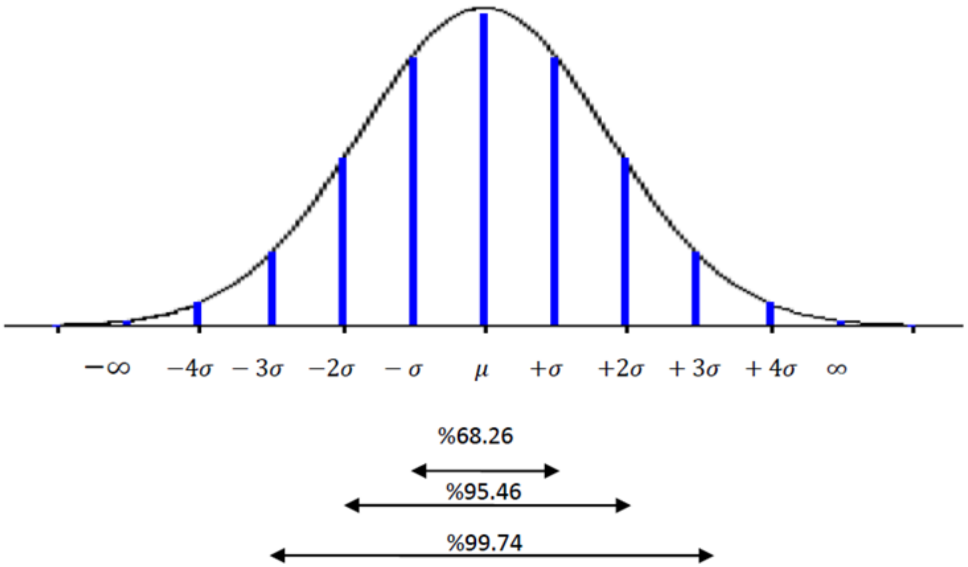
$$F(z) = P\left(Z \le z\right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} .e^{-\frac{1}{2}u^2} du \qquad \left[F(-\infty) = 0,\ \ F(+\infty) = 1\right]$$

$$\text{For } z = 0 \Rightarrow x = \mu \qquad \text{and } F(0) = 0{,}50$$

$$\text{For } z = 1 \Rightarrow x = \mu + 1.\sigma \qquad \text{and } F(1) = 0{,}8413$$

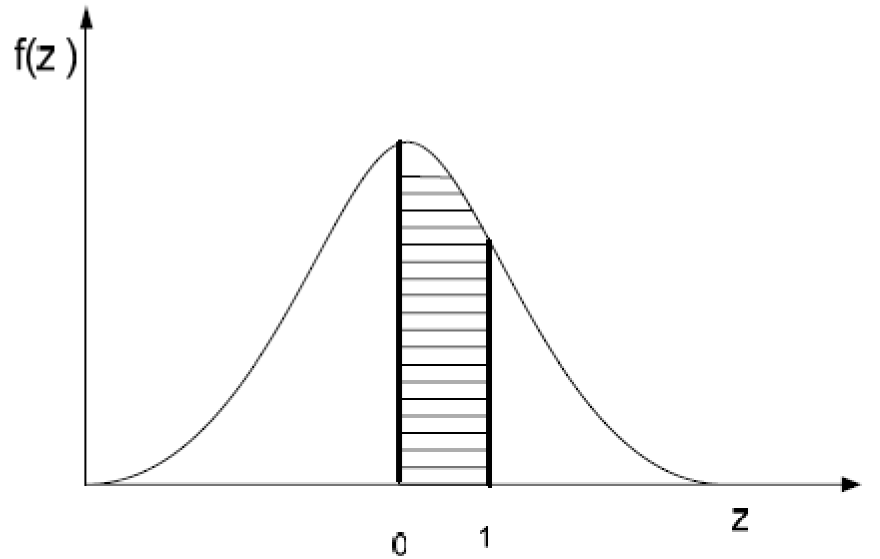
$$\text{For } z = 2 \Rightarrow x = \mu + 2.\sigma \qquad \text{and } F(2) = 0{,}9772$$

$$\text{For } z = 3 \Rightarrow x = \mu + 3.\sigma \qquad \text{and } F(3) = 0{,}9987$$



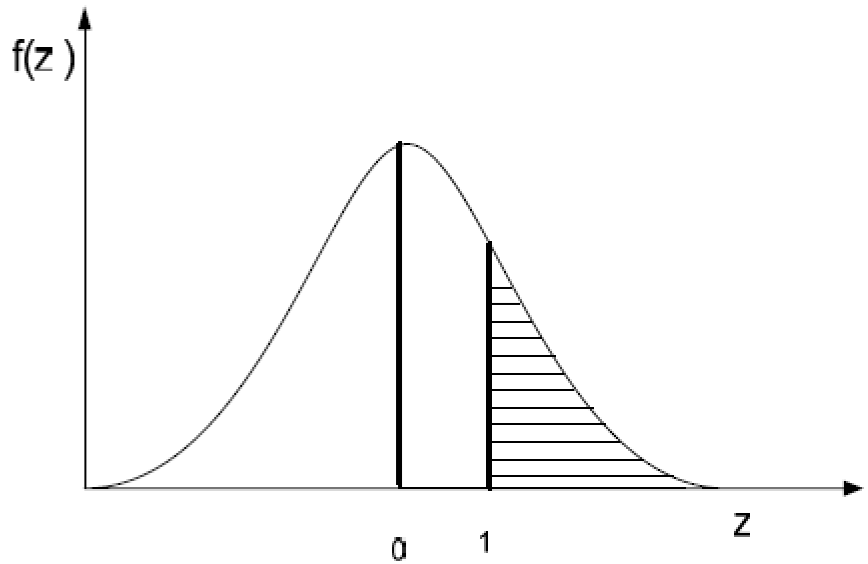
Probability calculation using the standard normal distribution table

$$P(0 < z < 1) = ?$$



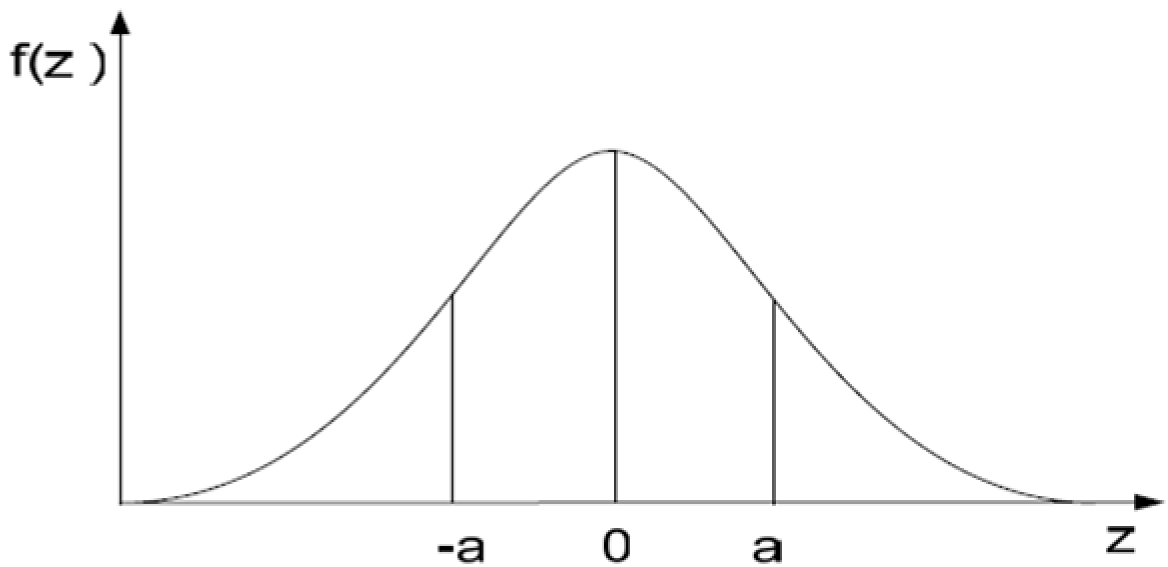
$$P(0 < z < 1) = 0{,}3413$$

$$P(z > 1) = ?$$

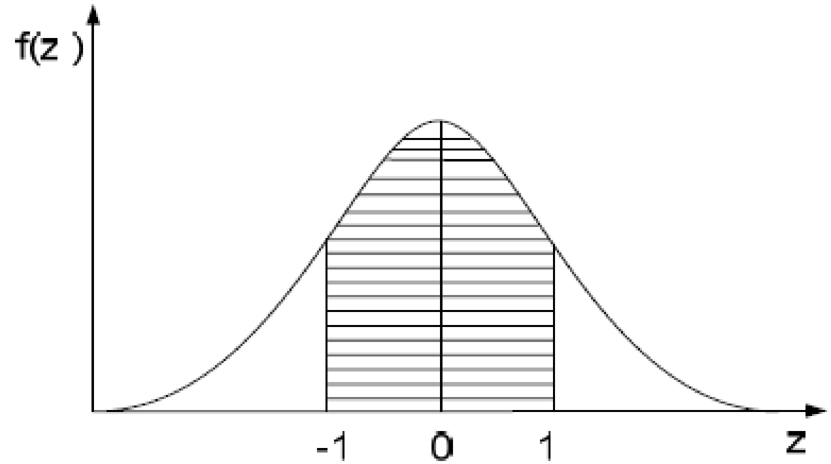


$$0,5 - \mathrm{P}(0 < z < 1) = 0,5 - 0,3413 = 0,1587$$

$$P(0 < z < a) = P(-a < z < 0)$$

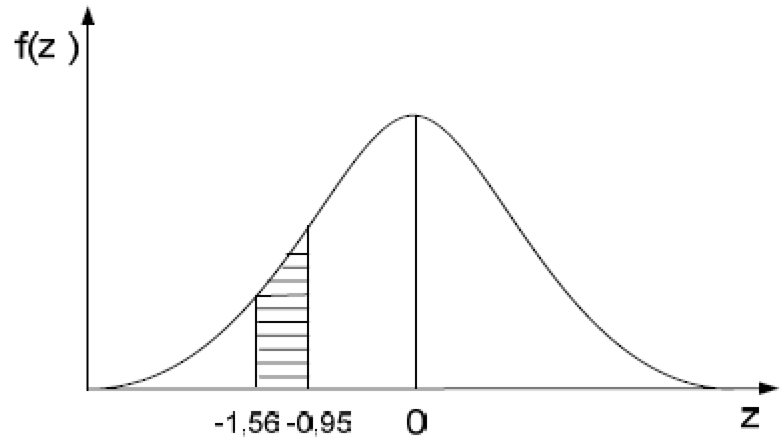


$$P(-1 < z < 1) = ?$$



$$P(-1 < z < 1) = P(-1 < z < 0) + P(0 < z < 1)$$
$$= 2 * P(0 < z < 1) = 2(0{,}3413) = 0{,}6826$$

$$P(-1{,}56 < z < -0{,}95) = ?$$



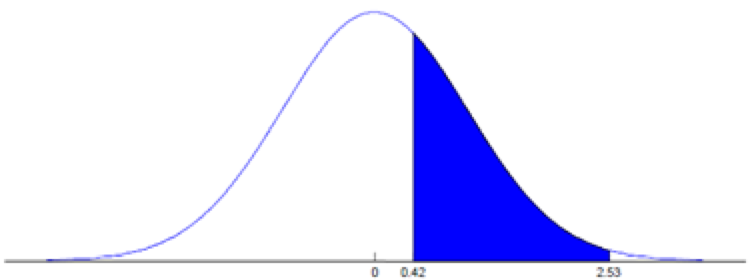
$$P(-1{,}56 < z < -0{,}95) = P(-1{,}56 < z < 0) - P(-0{,}95 < z < 0)$$

$$= 0{,}4406 - 0{,}3289 = 0{,}1117$$

Example: Calculate the probabilities given below for the standard normal distribution z variable.

a) $P(0.42 < Z < 2.53) =?$

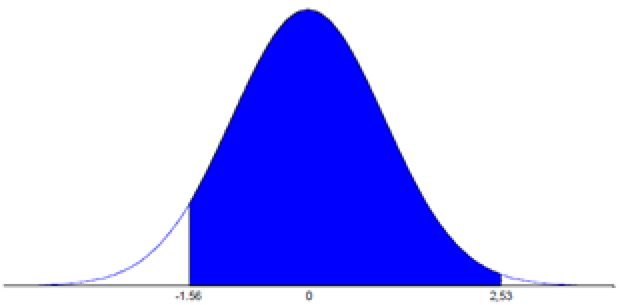
b) $P(-1.56 < Z < 2.53) =?$



$P(0.42 < Z < 2.53) = P(Z < 2.53) - P(Z < 0.42)$

=0.4943-0.1628
=0.3315

(a)



$P(-1.56 < Z < 2.53) =$ 0.4943+0.4406
=0.9349

(b)

Example: If X has a normal distribution with $\mu=3$ and $\sigma^2=4$, find $P(3<X<5)=?$

Example: The weight of the students in a class has a normal distribution with the mean $\mu=68.5$ kg and the standard deviation $\sigma=2.3$ kg.

a) What is the probability that a random chosen student in this class has weight of more than 72?
b) What percentage of the students have weight between 70 kg and 72 kg?

EXAMPLE

Find the probabilities that a random variable having the standard normal distribution will take on a value

(a) less than 1.72;

(b) less than –0.88;

(c) between 1.30 and 1.75;

(d) between –0.25 and 0.45.

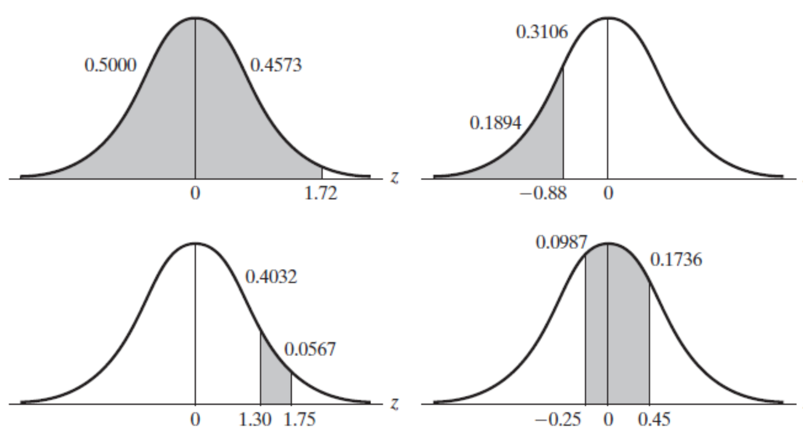
Solution

(a) We look up the entry corresponding to $z = 1.72$ in the standard normal distribution table, add 0.5000 and get $0.4573 + 0.5000 = 0.9573$.

(b) We look up the entry corresponding to $z = 0.88$ in the table, subtract it from 0.5000 and get $0.5000 - 0.3106 = 0.1894$.

(c) We look up the entries corresponding to $z = 1.75$ and $z = 1.30$ in the table, subtract the second from the first and get $0.4599 - 0.4032 = 0.0567$.

(d) We look up the entries corresponding to $z = 0.25$ and $z = 0.45$ in the table, add them and get $0.0987 + 0.1736 = 0.2723$.



EXAMPLE

With reference to the standard normal distribution table, find the values of z that correspond to entries of

(a) 0.3512;

(b) 0.2533.

Solution

(a) Since 0.3512 falls between 0.3508 and 0.3531, corresponding to $z = 1.04$ and $z = 1.05$, and since 0.3512 is closer to 0.3508 than 0.3531, we choose $z = 1.04$.

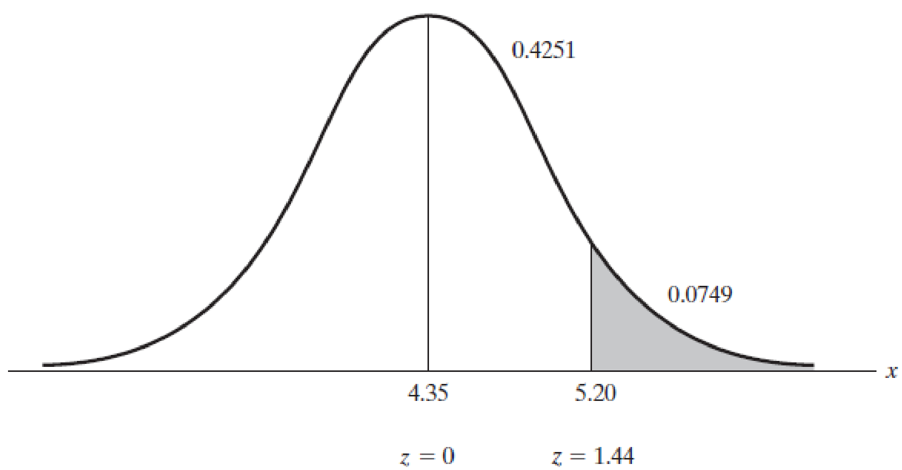
(b) Since 0.2533 falls midway between 0.2517 and 0.2549, corresponding to $z = 0.68$ and $z = 0.69$, we choose $z = 0.685$.

EXAMPLE

Suppose that the amount of cosmic radiation to which a person is exposed when flying by jet across the United States is a random variable having a normal distribution with a mean of 4.35 mrem and a standard deviation of 0.59 mrem. What is the probability that a person will be exposed to more than 5.20 mrem of cosmic radiation on such a flight?

Solution

Looking up the entry corresponding to $z = \dfrac{5.20 - 4.35}{0.59} = 1.44$ in the table and subtracting it from 0.5000 (see Figure) , we get $0.5000 - 0.4251 = 0.0749$.



Example: The weight of the tea boxes with capacity of 1 kg which are filled by machines has a normal distribution with the mean $\mu=1.03$ kg and the standard deviation $\sigma=0.02$ kg. What is the probability that a random tea box has weight of

a) Less than 1 kg?

b) More than 1.06 kg?

Example: Assume that the life of a certain type of batteries produced in a factory are normally distributed. If 33% of the batteries last less than 45 hours and 10% of the batteries last more than 65 hours, find the mean and the standard deviation of the distribution.

Example: An exam has a normal distribution with mean of 75 and the standard deviation of 8. The teacher gives A to the students whose grades are 90 or more. If 12 students take A, find the number of students attending to the exam.

THE NORMAL APPROXIMATION TO THE BINOMIAL DISTRIBUTION

If X has a Binomial distribution, we have

$$z = \frac{X - E(X)}{\sqrt{Var(X)}}$$

We can say that the random variable is distributed as standard normal for large values of n.

$$E(X) = np, \quad Var(X) = npq$$

$$z = \frac{X - np}{\sqrt{npq}} \sim N(0,1)$$

If the random variable of X has Binomial distribution, this variable's probability function is;

$$f(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0,1,\dots,n$$

When p is very small, with increasing the values of n, the binomial distribution approaches the poisson distribution,

If p is not too small and n is large, it becomes difficult to calculate the probabilities by using the binomial distribution.

$$X \sim N(np, npq)$$

in this case, the standard normal distribution can be used

However, since the binomial distribution is discrete and the normal distribution is continuous, correction should be made while converting to the standard normal distribution.

Correction of Continuity

Such correction is necessary, especially if n is not large enough. Because of its definition, the probability of realization of a single point in a continuous distribution is zero, and since it means the calculation of the probabilities expressed in a range, the probability of occurrence of a constant k is;

$$P(-0,5 \le k \le 0,5)$$

$$P(a \le X \le b) = P\left(\frac{a-np}{\sqrt{npq}} \le \frac{X-np}{\sqrt{npq}} \le \frac{b-np}{\sqrt{npq}}\right)$$

$$P(a \le X \le b) = P\left(\frac{a-np}{\sqrt{npq}} \le z \le \frac{b-np}{\sqrt{npq}}\right)$$

Taking into account the continuity correction in this inequality

$$P(a \le X \le b) = P\left(\frac{a-0,5-np}{\sqrt{npq}} \le z \le \frac{b+0,5-np}{\sqrt{npq}}\right)$$

This continuity correction to be made in probability calculations is given in the table below.

Probability	**Correction of Continuity**
$P(X = x)$	$P(x - 0.5 \leq X \leq x + 0.5)$
$P(X \leq x)$	$P(X \leq x + 0.5)$
$P(X < x)$	$P(X \leq x - 0.5)$
$P(X \geq x)$	$P(X \geq x - 0.5)$
$P(X > x)$	$P(X \geq x + 0.5)$
$P(a \leq X \leq b)$	$P(a - 0.5 \leq X \leq b + 0.5)$

Example: In studies related to alcohol dependence, the rate of alcohol dependence in children born from alcoholic parents was found to be 80%. 200 people were selected from 23-45 age groups. According to this, find the probabilities;

a) $P(X \geq 150)$

b) $P(X = 178)$

c) $P(158 < X < 161)$

d) $P(X < 190)$

$p: 0.80$

$np = 200 \times 0.80 = 160$

$npq = 200 \times 0.80 \times 0.20 = 32$

$So, \quad X \sim N(160{,}32)$

By using Binomial formula

a) $P(X \geq 150) = P(X = 150) + P(X = 151) + \cdots + P(X = 200)$

Calculation of this inequality is very difficult. With the approach to normal distribution,results can be achieved with fewer operations using the table of standard normal distribution.

$$= P(X \geq 149.5)$$

$$= \mathrm{P}\left(Z \geq \frac{149.5 - 160}{\sqrt{32}}\right)$$

$$= \mathrm{P}(Z \geq -1.84)$$

$$= 0.5 + 0.4671$$

$$= 0.9671$$

b) $P(X = 178) = P(177.5 \leq X \leq 178.5)$

$$= P\left(\frac{177.5-160}{5.7} \leq Z \leq \frac{178.5-160}{5.7}\right)$$

$$= P(3.07 \leq Z \leq 3.25)$$

$$= 0.4994 - 0.4989$$

$$= 0.0005$$

c) $P(158 < X < 161) = P(158.5 \leq X \leq 160.5)$

$$= P\left(\frac{158.5 - 160}{5.7} \leq Z \leq \frac{160.5 - 160}{5.7}\right)$$

$$= P(-0.26 \leq Z \leq 0.09)$$

$$= 0.0359 + 0.1026$$

$$= 0.1385$$

d) $P(X < 190) = P(X \leq 189.5)$

$$= P\left(Z \leq \frac{189.5 - 160}{5.7}\right)$$

$$= P(Z \leq 5.16)$$

$$= 1$$

Example: A drug is effective in 60% of patients. What is the probability that the number of patients who do not recover when 30 patients are tried to cure with this drug will be less than 6?

Solution:

Here Normal approximation to the Binomial distribution will be used

X: number of patients not recovering

$$p: 0.40$$

$$np = 30 \times 0.40 = 12$$

$$npq = 30 \times 0.40 \times 0.60 = 7.2$$

$$X \sim N(12, 7.2)$$

$$P(X < 6) = P\left(Z < \frac{(6 - 0.5) - 12}{\sqrt{7.2}}\right) = P(Z < -2.42)$$

$$= 0.5 - 0.4922$$

$$= 0.0078$$